

mation from reticle seekers. A small IFOV ($\approx$ two milliradian) is scanned in a preset pattern and spatial information is used to determine the time instant of appearance of the target within the field-of-view. Detectors are therefore activated only within the time gate around the predicted time. This allows the missile to avoid large clutter and false targets outside the time gate. It also makes seeker highly immune to IR countermeasures.

Imaging seekers use an array of detectors called imaging focal plane

arrays (Fig. 10) instead of reticles to build an image of the scene in front (Fig. 11). The image may be created by scanning the scene and using a linear array or a two-dimensional staring array. Imaging seekers are very expensive, require huge processing power and complex tracking algorithms. These are therefore employed only under demanding operational requirements. Reticle seekers, on the other hand, are less expensive and easy to manufacture and operate, and have a proven reliability and accuracy.

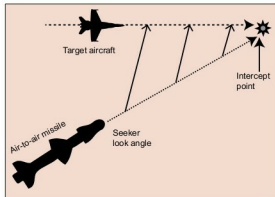


Fig. 9: Proportional navigation



Fig. 10: Imaging focal plane arrays

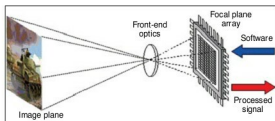


Fig. 11: Imaging IR seeker concept



Fig. 12: FGM-148 Javelin

IR-guided missile systems

Some of the better known IR-guided missile systems are briefly described in the following paragraphs. These missile systems have been grouped into three categories, namely, anti-tank-guided missiles (ATGMs), surface-to-air missiles (SAMs) and air-to-air missiles (AAMs).

Anti-tank-guided missiles. ATGMs, also known as anti-armour missiles or anti-tank missiles, are designed to precisely hit and destroy heavily-armoured military vehicles including all combat and transportation vehicles. Different variants of anti-tank missile systems include shoulder-fired weapons, large-tripod-mounted weapons, vehicle-mounted weapons and systems launched from airborne launchers. American Javelin, German PARS-3 LR, Israeli Spike and Indian Nag are some examples.

American FGM-148 Javelin (Fig. 12) is a man-portable third-generation fire-and-forget anti-tank guide missile jointly developed by Raytheon and Lockheed Martin. It has an effective firing

range of 75m to 2500m, with 4750m being the maximum range. It uses a tandem-shaped charge warhead that can penetrate reactive armour. The missile can be used in both top attack mode to hit usually thin-top armour of the target vehicle and direct mode to hit buildings and airborne targets. It uses an IIR seeker and on-board tracker to make it a fire-and-forget missile. It was introduced into service in 1996 and is in service till date. It was successfully used in Operation Enduring Freedom (war in Afghanistan) and Operation Iraqi Freedom.

PARS 3 LR (Fig. 13) is an autonomous fire-and-forget missile intended for long-range applications and designed to destroy ground (tanks and armoured vehicles), air (helicopters) and other individual targets. Manufactured by Parsys GmbH, MBDA Deutschland GmbH and Diehl BGT Defence, PARS 3 LR is also known as TRIGAT-LR and AC-3G. The missile can be launched from a ground vehicle or a helicopter and can be fired in salvos of up to four missiles in eight seconds. The missile has a specified operational range of 500m to 5000m, which is extendible up to 7000m. It can be used in both top (terminal dive) mode as well as direct mode. Fig. 14 (top part) shows the flight trajectory for top-attack mode employed for anti-armour role with the missile launched from either a land vehicle or a helicopter.

It also shows the flight trajectory for direct-attack mode (bottom part) employed for an anti-helicopter role with the missile launched from either a land vehicle or a helicopter. The missile uses a passive IR seeker that locks on to the target before the missile is fired. It uses tandem-shaped charge for maximum lethality against modern reactive armour. German army has authorised series production of the missile system. The delivery began in mid-2013 and shall continue till 2016.

Spike is a fourth-generation man-portable fire-and-forget anti-tank and anti-personnel missile designed and developed by Rafael Advanced Defence Systems. The missile can be



Fig. 13: PARS 3 LR